# Analyyysi I: Limits

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**Abstract**

Notes for the University of Helsinki course MAT11003, taught in Finnish from 01.08.2026–17.10.2026.

# Contents



# 1 Course information

The official course information is available on the University of Helsinki course page.

# 2 Lecture notes

## 2.1 01.09.2026

### 2.1.1 The absolute value function

**Definition 2.1.** For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of $x$, denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points $x$ and $y$ on the real line.

**Example 2.2.** Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x-y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y-h, y+h). \end{aligned}$$

Thus, $|x - y| < h$ means that $x$ lies in the open interval of radius $h$ centred at $y$.

**Example 2.3.** As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x-1| < 3 &\iff -3 < 2x - 1 < 3 \\ &\iff -2 < 2x < 4 \\ &\iff -1 < x < 2 \\ &\iff x \in (-1, 2). \end{aligned}$$

**Theorem 2.4** (Triangle and reverse triangle inequalities)**.** *For all $x, y \in \mathbb{R}$,*

$$\begin{aligned} |x+y| &\le |x| + |y|, \\ ||x| - |y|| &\le |x-y|. \end{aligned}$$

*Proof.* By definition 2.1,

$$\begin{aligned} -|x| &\le x \le |x|, \\ -|y| &\le y \le |y|. \end{aligned}$$

Adding the inequalities gives

$$-\Big(|x| + |y|\Big) \le x + y \le |x| + |y|,$$

which is equivalent to $|x + y| \le |x| + |y|$.

Applying this triangle inequality to $x = (x - y) + y$ gives

$$|x| - |y| \le |x - y|.$$

Interchanging $x$ and $y$ similarly gives $|y| - |x| \le |x - y|$. Therefore

$$-|x-y| \le |x| - |y| \le |x-y|,$$

which is equivalent to the reverse triangle inequality. □

## 2.2 03.09.2026

### 2.2.1 More on absolute values

**Theorem 2.5.** *We have*

$$|xy| = |x||y|$$

*for all $x, y \in \mathbb{R}$.*

*Proof.* We prove by splitting into four cases. Firstly, if $x, y \geq 0$, we have $xy \geq 0$, so $|xy| = xy$. But in this case also both $|x| = x$ and $|y| = y$, so $|xy| = xy = |x||y|$. Secondly, if $x, y < 0$, we have $xy > 0$, $|xy| = xy = (-x)(-y) = |x||y|$. Now suppose $x < 0$ and $y \geq 0$. Then $xy \leq 0$, and hence $|xy| = -xy = (-x)y = |x||y|$. The case $x \geq 0$, $y < 0$ follows from commutativity of multiplication of real numbers. $\square$

**Example 2.6.** Let $x \in (1, 3)$ and $y \in (4, 8)$. Find an upper bound for

$$|x + y - 8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x + y - 8| &= |(x - 2) + (y - 6)| \\ &\leq |x - 2| + |y - 6| \\ &< 1 + 2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x + y - 8|$.

**Example 2.7.** Let $x \in (2, 6)$ and $y \in (2, 8)$. Find an upper boud for

$$|xy - 20|.$$

We have $|x - 4| < 2$ and $|y - 5| < 3$. We have

$$\begin{aligned} |xy - 20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x - 4| + 4|y - 5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

**Example 2.8.** Find a positive real number $m$ such that

$$|x^2 - 16| \leq m|x - 4|,$$

where $x \in (0, 10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x + 4| \leq m.$$

Since $x \in (0, 10)$, $|x + 4| \in (4, 14)$. We have thus obtained

$$|x + 4| \leq 14,$$

so we can choose $m = 14$.

### 2.2.2 Field and ordering properties of $\mathbb{R}$

**Definition 2.9.** $(F, +, \cdot)$ is a field if and only if

1. The algebraic structure $(F, +)$ is an *abelian group.*
2. The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
3. The operation $\cdot$ distributes over $+$.

**Theorem 2.10.** $\mathbb{R}$ *is a field.*

*Proof.* Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y + z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. □

**Definition 2.11** (Ordered field)**.** An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

1. exactly one of $x < y$, $x = y$, or $x > y$ holds;
2. if $x < y$ and $y < z$, then $x < z$;
3. if $x < y$, then $x + z < y + z$; and
4. if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

**Theorem 2.12.** *The field* $(\mathbb{R}, +, \cdot)$*, equipped with the usual order* $<$*, is an ordered field.*

*Proof.* We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. □

**Theorem 2.13** (Uniqueness of additive identity). *The identity element $0_F$ of the abelian group $(F, +)$ is unique.*

*Proof.* Suppose that $0_F$ and $0'_F$ are both additive identity elements of $(F, +)$. Since $0_F$ is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. $\square$

**Theorem 2.14** (Cancellation law). *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

*Proof.* Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

$\square$

## 2.3 08.09.2026

### 2.3.1 Sequences

**Definition 2.15.** Let $X$ and $Y$ be sets. A function $f\colon X \to Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique $y$ is denoted by $f(x)$.

The set $X$ is the *domain* of $f$, and $Y$ is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

**Definition 2.16.** A sequence of real numbers is a function $x\colon \mathbb{N} \to \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by $x_n$, is the $n$-th term of the sequence. We denote the sequence by $(x_n)_{n\in\mathbb{N}}$, or simply by $(x_n)$.

### 2.3.2 Limit of a sequence

**Definition 2.17.** Let $(x_n)$ be a sequence of real numbers. We say that $(x_n)$ *converges* to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \epsilon.$$

In this case, we write $x_n \to a$ as $n \to \infty$, or $\lim_{n\to\infty} x_n = a$.

Let us look at some examples.

**Example 2.18.** Let $(x_n)$ be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n\to\infty} x_n = 2.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\epsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= |\frac{2n}{n+1} - 2| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\epsilon}} = \epsilon. \end{aligned}$$

Thus $x_n \to 2$ when $n \to \infty$.

**Example 2.19.** Let $(x_n)$ be defined by

$$\begin{aligned} x_0 &= \cdots = x_7 = 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n\to\infty} x_n = 7.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max\left\{8, \frac{1}{\epsilon}\right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \epsilon.$$

Thus $x_n \to 7$ as $n \to \infty$.

Let us now tackle a simple divergence proof.

**Example 2.20.** Let us show that the sequence $x_n$ defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \epsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max\{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \le |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \ge n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore $(x_n)$ diverges.

**Example 2.21.** Let $(x_n)$ be defined by

$$\begin{aligned} x_0 &= \cdots = x_{42} = 1, \\ x_n &= \frac{n}{n-42}, \quad n \ge 43. \end{aligned}$$

We will show that

$$\lim_{n\to\infty} x_n = 1.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max\left\{84, \frac{84}{\epsilon}\right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left|\frac{n}{n-42} - 1\right| \\ &= \frac{42}{n-42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \epsilon. \end{aligned}$$

Thus $x_n \to 1$ as $n \to \infty$.